

Numerical Results for the Bermudan Put Experiments

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This report presents the fixed Heston benchmark, the current default gDMR case, the equal-delta gDMR case, and the strike tests carried out under the equal-delta setup.

Fixed Heston benchmark. This case is kept to compare the implementation against the published Heston Bermudan-put benchmark.

Table 1: Heston benchmark parameters.

Quantity	Value
Model	Heston
Option	Bermudan put
S_0	10
K	10
T	1
r	0.02
v_0	0.15
κ	5
θ	0.16
η	0.9
ρ	0.1
Exercise dates	12
LSMC paths	500,000
LSMC steps M	1,000
LSMC basis degree	3
Hybrid paths N	10,000
Hybrid low paths N_{low}	10,000
Hybrid asset grid points N_S	512
Hybrid basis degree	3
Finite-difference reference	1.4507

Table 2: Heston benchmark results.

Estimator	Published value	Run value	Run – FD	Run – published
LSMC direct	1.4494	1.458896	0.008196	+0.009496
LSMC low	1.4487	1.452544	0.001844	+0.003844
Hybrid direct	1.4530	1.452988	0.002288	-0.000012
Hybrid low	1.4529	1.445493	0.005207	-0.007407

Default gDMR case ($\delta_1 = \delta_2 = 0.50$). This is the current tuned default setup. Relative errors in the next table are measured against the LSMC direct estimator.

Table 3: Default gDMR parameters.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100
K	100
T	1
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
ρ_{12}	0.20
ρ_{13}	0.10
ρ_{23}	0.10
δ_1	0.50
δ_2	0.50
Exercise dates N_{ex}	100
Euler steps M	600
LSMC paths	1,000,000
LSMC basis degree	3
LSMC basis size	16
Hybrid paths N	30,000
Hybrid low paths N_{low}	30,000
Hybrid asset grid points N_S	181
Hybrid Hermite nodes N_{hermite}	64
Hybrid volatility basis degree	3
Hybrid volatility basis size	10

Table 4: Default gDMR results.

Estimator	Price	Standard error	Relative error
LSMC direct	6.384535	0.007943	reference
LSMC low	6.387249	0.007939	0.04% vs LSMC direct
Hybrid direct	6.421758	0.000652	0.58% vs LSMC direct
Hybrid low	6.265070	0.045711	1.87% vs LSMC direct

Equal-delta gDMR case ($\delta_1 = \delta_2 = 0.75$). This run keeps the same tuned numerical setup as the default gDMR case and changes only the two elasticity parameters.

Table 5: Equal-delta gDMR parameters.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100
K	100
T	1
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
ρ_{12}	0.20
ρ_{13}	0.10
ρ_{23}	0.10
δ_1	0.75
δ_2	0.75
Exercise dates N_{ex}	100
Euler steps M	600
LSMC paths	1,000,000
Hybrid paths N	30,000
Hybrid low paths N_{low}	30,000
Hybrid asset grid points N_S	181
Hybrid Hermite nodes N_{hermite}	64

Table 6: Equal-delta gDMR results.

Estimator	Price	Standard error	Relative error
LSMC direct	6.653560	0.007822	reference
LSMC low	6.655062	0.007819	0.02% vs LSMC direct
Hybrid direct	6.696152	0.000375	0.64% vs LSMC direct
Hybrid low	6.545257	0.045080	1.63% vs LSMC direct

Strike tests under the equal-delta setup. All model parameters and numerical settings are the same as in Table 4, except for the strike reported below.

Table 7: Strike-test parameters under the equal-delta setup.

Case	S_0	K
ITM put	100	110
ATM put	100	100
OTM put	100	90

Table 8: Strike-test results under the equal-delta setup.

Case	Strike	LSMC direct	LSMC low	Hybrid direct	Hybrid low	Hybrid direct rel. error	Hybrid low rel. vs LSMC direct
ITM put	110	12.735850	12.734082	12.767870	12.634384	0.25%	0.80%
ATM put	100	6.653560	6.655062	6.696152	6.545257	0.64%	1.63%
OTM put	90	2.743578	2.734472	2.791426	2.668242	1.74%	2.75%